Lesson 1: Rational Expressions and Non-Permissible Values

Recognizing rational expressions, evaluating them, and finding the values of the variable that make them undefined.

1. What Is a Rational Expression?

A **rational number** is a quotient $\frac{p}{q}$ of two integers with $q \neq 0$. In the same way, a **rational expression** is a quotient $\frac{P}{Q}$ of two **polynomials**, where $Q \neq 0$. A polynomial has only whole-number exponents on its variables: no variable under a radical, in an exponent or in a denominator. A polynomial by itself, such as $x^2 - 4x + 1$, is also a rational expression (its denominator is 1).

Rational expression:	$\frac{\text{polynomial}}{\text{polynomial}}$, the denominator not 0.	Examples: $\frac{5}{x}$, $\frac{x+3}{x^2-4}$, $\frac{2a^2-1}{7}$.

To **evaluate** a rational expression, substitute the value and simplify the numerator and the denominator separately. A numerator of 0 is fine (the value is 0); a denominator of 0 makes the expression **undefined**.

Example 1

Is each a rational expression? Explain.

(a) $\frac{x^2-5}{3x}$ (b) $\frac{4}{\sqrt{x}+1}$ (c) $\frac{2y+7}{9}$ (d) $\frac{x}{x^2+1}$

Example 2

Evaluate $\frac{x^2-4x}{2x+6}$ for each value of x.

(a) $x = 5$ (b) $x = -1$ (c) $x = 4$ (d) $x = -3$

2. Why Division by Zero Is Undefined

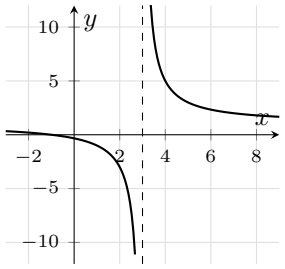
Division undoes multiplication: $\frac{12}{3} = 4$ because $4 \times 3 = 12$. If $\frac{12}{0}$ had a value k, then $k \times 0$ would be 12, but $k \times 0 = 0$ for every k. So $\frac{12}{0}$ has no value. ($\frac{0}{0}$ fails too: every k works, so there is no single value.) A table of values shows what happens near a value that makes a denominator 0: the expression grows without bound and changes sign, so no number fits there. On a calculator, enter the expression as Y1 and look at its TABLE: it shows ERROR at that value.

Example 3

(a) Complete the table for $\dfrac{x+1}{x-3}$.

x	2.9	2.99	3	3.01	3.1	4
$\dfrac{x+1}{x-3}$						

(b) Describe what happens to the value as x gets close to 3, and explain why there is no value at $x = 3$.



The graph of $y = \dfrac{x+1}{x-3}$ has no point at $x = 3$.

3. Non-Permissible Values

A **non-permissible value** (NPV) is a value of the variable that makes a denominator equal 0. These values are the **restrictions** on the variable. Only the denominator matters: a numerator equal to 0 is allowed.

To find the NPVs: set each denominator (each factor with a variable) equal to 0 and solve. Write them as $x \neq 3$, or $x \neq -2, 5$, or $x \neq \pm 4$.

A denominator with no variable, such as 8, gives no NPV. A denominator like $x^2 + 16$ is never 0 (a square is never negative, so $x^2 + 16 \geq 16$), so it gives no NPV either.

Example 4

Determine the non-permissible values.

(a) $\dfrac{9}{5x}$ (b) $\dfrac{2y-1}{y+8}$ (c) $\dfrac{m^2+4}{12-3m}$

(d) $\dfrac{x}{7x+2}$ (e) $\dfrac{4k+3}{10}$ (f) $\dfrac{3x}{x^2+16}$

4. Factor the Denominator First

When a denominator is a trinomial, a difference of squares or has a common factor, **factor it fully** (Unit 4) and use the zero product property: each factor with a variable gives an NPV. Ignore the numerator, even if it shares a factor with the denominator. In an expression with more than one fraction, **every** denominator gives NPVs. With two variables, solve for the variable asked for (for example, x in terms of y).

Example 5

Determine the non-permissible values.

(a) $\dfrac{x-4}{x^2+2x-15}$ (b) $\dfrac{8}{x^2-36}$ (c) $\dfrac{n+2}{3n^2-12n}$

(d) $\dfrac{1}{2x^2+7x-4}$ (e) $\dfrac{y+3}{y^2-10y+25}$ (f) $\dfrac{5}{x^3-9x}$

Example 6

Determine the non-permissible values (in (b), for x in terms of y).

(a) $\dfrac{5}{3a^2b}$ (b) $\dfrac{x+y}{x^2-25y^2}$ (c) $\dfrac{x}{x-1}+\dfrac{2}{x^2+4x}$

5. Same Values, Different Restrictions

Two rational expressions can give the same value for almost every x but have different NPVs. This is the idea behind simplifying (Lesson 2): the restrictions of the original expression must be kept.

Example 7

(a) Complete the table. Write "undefined" where there is no value.

x	-4	-3	-2	0	1	2
$\dfrac{3x+6}{x^2+5x+6}$						
$\dfrac{3}{x+3}$						

(b) For which value of x do the expressions differ? (c) State the NPVs of each expression.

6. Building Expressions and Using Them in Context

To build an expression with given NPVs, work backwards as in Unit 4: an NPV $x = r$ comes from a factor $(x - r)$ in the denominator, and $x = \frac{p}{q}$ from $(qx - p)$. In a context, the NPVs are values the variable can never take, and the situation often restricts the variable further (a length or a speed must be positive, a count must be a whole number).

Example 8

Write a rational expression with a numerator of 1 and the denominator expanded that has the given NPVs.

(a) $x \neq -1, 6$ (b) $x \neq 0, \frac{2}{3}$ (c) $x \neq \pm 4$ (d) no NPVs

(e) The expression $\dfrac{1}{x^2 + kx - 12}$ has an NPV of $x = 2$. Determine k and the other NPV.

Example 9

A boat travels 36 km up a river against a current of 3 km/h. If its speed in still water is v km/h, its speed upstream is $(v - 3)$ km/h and the trip takes $t = \dfrac{36}{v - 3}$ hours.

(a) Determine the time for $v = 15$ and for $v = 12$.

(b) State the NPV. What would it mean for the boat?

(c) Which values of v make sense here?

Lesson 2: Simplifying Rational Expressions by Factoring

Writing rational expressions in simplest form by dividing out common factors, with the non-permissible values stated first.

1. Equivalent Forms and Simplest Form

A fraction is simplified by dividing out a common factor: $\dfrac{18}{24} = \dfrac{3 \cdot 6}{4 \cdot 6} = \dfrac{3}{4}$. Rational expressions work the same way. Two expressions are **equivalent** if they have the same value for every permissible value of the variable. An expression is in **simplest form** when its numerator and denominator have no common factor other than 1 (or -1).

To simplify a rational expression: (1) factor the numerator and the denominator fully; (2) state the NPVs from the **original** denominator; (3) divide out the common factors; (4) write the result with its NPVs.
The simplified expression keeps the NPVs of the original, even when the factor that caused them is gone.

With monomials, use the exponent laws: $\dfrac{x^5}{x^2} = x^3$.

Example 1

Simplify. State the non-permissible values.

(a) $\dfrac{18x^5}{24x^2}$

(b) $\dfrac{-10a^2b}{35ab^3}$

(c) $\dfrac{6(x-2)}{9(x-2)(x+1)}$

(d) $\dfrac{(m+4)^2}{(m+4)(m-3)}$

2. Divide Out Factors, Not Terms

Only a **factor** of the whole numerator and of the whole denominator can be divided out. A **term** (a part joined by + or −) cannot. For example, $\dfrac{x+6}{x+2}$ is not $\dfrac{6}{2}=3$: try $x=2$, which gives $\dfrac{8}{4}=2$, not 3. To check a simplification, substitute a permissible value into the original and the result: they must agree.

Example 2

Each simplification is wrong. Explain the error and give the correct simplest form with its NPVs.

(a) $\dfrac{x+8}{4}=x+2$

(b) $\dfrac{3x+12}{3}=x+12$

(c) $\dfrac{x^2-25}{x-5}=x-5$ (the squares were "cancelled")

(d) $\dfrac{x^2+4x}{x}=x+4$

3. Factoring the Numerator and the Denominator

Use every factoring method from Unit 4: common factor first, then a difference of squares or a trinomial (x^2+bx+c or decomposition for ax^2+bx+c). Factor fully before dividing anything out.

Example 3

Simplify. State the non-permissible values.

(a) $\dfrac{4x-20}{x^2-25}$

(b) $\dfrac{x^2+2x-24}{x^2-7x+12}$

(c) $\dfrac{3y^2-12y}{6y^2-18y-24}$

(d) $\dfrac{2x^2+7x-15}{4x^2-9}$

Example 4

Simplify. State the non-permissible values.

(a) $\dfrac{6a^2+13a+6}{3a^2-4a-4}$

(b) $\dfrac{x^3-4x}{x^3+x^2-6x}$

4. Opposite Binomials

$a-b$ and $b-a$ are **opposites**: $b-a=-(a-b)$. For example, $7-x=-(x-7)$. So a quotient of opposites is -1. To use this, factor -1 out of one of them (also out of a trinomial with a negative x^2 term: $12+x-x^2=-(x^2-x-12)$).

$\dfrac{a-b}{b-a}=\dfrac{a-b}{-(a-b)}=-1,\ \ a\neq b.$ But $\dfrac{a+b}{b+a}=1$: $3+x$ and $x+3$ are the same, not opposites.

Example 5

Simplify. State the non-permissible values.

(a) $\dfrac{x-7}{7-x}$

(b) $\dfrac{10-5a}{a^2-4}$

(c) $\dfrac{9-x^2}{x^2+x-12}$

(d) $\dfrac{12+x-x^2}{2x^2-7x-4}$

5. Two Variables and Equivalent Forms

With two variables, factor in the same way and state the NPVs for the variable asked for (for example, x in terms of y). To decide whether two expressions are equivalent, simplify both, or rewrite one to match the other. To write an equivalent expression with a new denominator, multiply the numerator and the denominator by the same factor.

Example 6

Simplify, and state the non-permissible values for x in terms of y.

(a) $\dfrac{x^2-9y^2}{2x+6y}$

(b) $\dfrac{x^2-xy-6y^2}{3y-x}$

(c) $\dfrac{2x^2+5xy-3y^2}{4x^2-y^2}$

Example 7

Is each expression equivalent to $\dfrac{x-3}{x+2}$ (for its permissible values)? Explain.

(a) $\dfrac{x^2-9}{x^2+5x+6}$

(b) $\dfrac{3-x}{-x-2}$

(c) $\dfrac{3-x}{x+2}$

(d) Determine the missing numerator: $\dfrac{x-3}{x+2}=\dfrac{?}{x^2-4}$.

(e) The expression $\dfrac{x^2+kx-15}{x^2-9}$ simplifies to $\dfrac{x+5}{x+3}$. Determine k.

6. Rational Expressions in Geometry

A missing length is often a quotient: length $= \dfrac{\text{area}}{\text{width}}$. Simplify the quotient by factoring, then use it. In a context, the variable must also keep every length positive.

Example 8

A rectangular garden has an area of $(2x^2+9x-5)$ m^2 and a length of $(x+5)$ m.

(a) Write and simplify an expression for its width.

(b) For which value of x is the garden a square? State its side length.

(c) Determine the exact perimeter and area when $x = 3+\sqrt{2}$.

Example 9

A box (rectangular prism) has a square base of side x cm and a height of $(x+3)$ cm.

(a) Write the surface area and the volume in factored form.

(b) Write the ratio $\dfrac{SA}{V}$ in simplest form and state the NPVs.

(c) Evaluate the ratio for $x = 3$ and $x = 6$. What do you notice?

Lesson 3: Multiplying Rational Expressions

Factoring every numerator and denominator, stating the non-permissible values, dividing out common factors and multiplying what is left.

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1. Multiplying Fractions and Monomials

To multiply fractions, multiply the numerators and multiply the denominators. It is easier to **divide out common factors first** (a factor of *any* numerator with a factor of *any* denominator), so the numbers stay small and the answer is already in simplest form. Rational expressions are multiplied the same way.

$\frac{A}{B} \cdot \frac{C}{D} = \frac{AC}{BD}, \quad B \neq 0, D \neq 0.$	Divide out common factors before multiplying.

For monomials, use the exponent laws ($\frac{x^5}{x^2} = x^3$). A variable is restricted only if it appears in a **denominator**: each such variable $\neq 0$.

Example 1

Simplify. State the non-permissible values where there are variables.

(a) $\frac{14}{15} \times \frac{25}{21}$

(b) $\frac{9}{16} \times \left(-\frac{20}{27}\right)$

(c) $\frac{6x^3}{5y} \cdot \frac{10y^4}{9x}$

(d) $\frac{-4a^2b}{15c^3} \cdot \frac{25bc}{6a^5}$

2. Multiplying Rational Expressions with Polynomials

With polynomials, you can only divide out common **factors**, so factor first. A polynomial such as $x^2 - 9$ on its own is a rational expression with denominator 1.

To multiply rational expressions: (1) factor every numerator and denominator fully; (2) state the NPVs: every value that makes *any* original denominator 0; (3) divide out factors common to any numerator and any denominator; (4) multiply what is left. Leave the answer in factored form.

A factor that is divided out still gives an NPV: the original product is undefined there, even though the simplified answer is not.

Example 2

Simplify. State the non-permissible values.

(a) $\frac{x+3}{4x} \cdot \frac{10x^2}{(x+3)(x-1)}$

(b) $\frac{5x-15}{x^2+2x} \cdot \frac{3x+6}{x^2-9}$

(c) $\frac{8y^2}{y^2-25} \cdot \frac{2y+10}{12y}$

Example 3

Simplify. State the non-permissible values.

(a) $\dfrac{x^2 - x - 12}{x^2 + 5x + 6} \cdot \dfrac{x^2 - 4}{x^2 - 6x + 8}$

(b) $\dfrac{2a^2 + 7a - 4}{a^2 - 16} \cdot \dfrac{a^2 - 3a - 4}{4a^2 - 1}$

(c) $\dfrac{3x^2 - 12x}{6x^2 + x - 2} \cdot \dfrac{4x^2 - 1}{x^2 - 8x + 16}$

Calculator check. Enter the original product as Y_1 and your answer as Y_2 and look at the TABLE: the two columns agree for every x except the NPVs, where Y_1 shows ERROR.

3. Opposite Factors

From Lesson 2, $\dfrac{a - b}{b - a} = -1$. When a factor is the opposite of another, such as $(7 - x)$ and $(x - 7)$, write it as $-(x - 7)$ so that the two can divide out, leaving -1. Also factor out -1 (or a negative common factor) when the leading coefficient is negative: $9 - a^2 = -(a^2 - 9) = -(a - 3)(a + 3)$.

Example 4

Simplify. State the non-permissible values.

(a) $\dfrac{x - 7}{x^2 + 3x} \cdot \dfrac{4x + 12}{14 - 2x}$

(b) $\dfrac{9 - a^2}{a^2 + a - 6} \cdot \dfrac{a^2 - 4a + 4}{2a^2 - 6a}$

4. Products of Three Expressions and Two Variables

The same steps work for any number of factors: factor all of them, take the NPVs from every denominator, and divide out across the whole product. With two variables, a factor such as $x - 3y$ in a denominator gives the restriction $x \neq 3y$; a factor y gives $y \neq 0$.

Example 5

Simplify. State the non-permissible values.

(a) $\dfrac{x^2-4}{2x+6}\cdot\dfrac{x^2+7x+12}{x^2+2x}\cdot\dfrac{4x^2}{x^2+2x-8}$

(b) $\dfrac{x^2-9y^2}{x^2+xy-2y^2}\cdot\dfrac{x^2-xy}{2x-6y}$

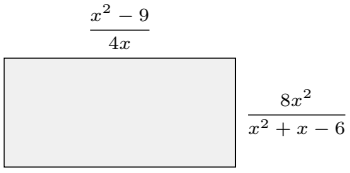
5. Problems with Products

Areas and volumes multiply lengths, so a shape whose dimensions are rational expressions has a product of rational expressions as its area. In a context, the variable must also make every length positive.

Example 6

A rectangle has length $\dfrac{x^2-9}{4x}$ cm and width $\dfrac{8x^2}{x^2+x-6}$ cm, where $x>3$.

(a) Write and simplify an expression for its area.



(b) Determine the exact area when $x=2\sqrt{3}$, in simplest form. (Use Unit 2: rationalize the denominator.)

Lesson 4: Dividing Rational Expressions

Multiplying by the reciprocal of the divisor, with non-permissible values from the denominators and from the divisor's numerator.

1. Dividing by a Fraction

Dividing by a number is the same as multiplying by its **reciprocal**: $\div\dfrac{3}{4}$ is the same as $\times\dfrac{4}{3}$. Only the expression right after the $\div$ sign (the **divisor**) is turned upside down. After that, it is a multiplication, done as in Lesson 3.

$\dfrac{A}{B}\div\dfrac{C}{D}=\dfrac{A}{B}\cdot\dfrac{D}{C}$	(keep the first expression, change $\div$ to $\times$, flip the divisor)

Example 1

Divide. Write each answer in lowest terms.

(a) $\frac{8}{15} \div \frac{12}{25}$ (b) $-\frac{9}{14} \div 6$ (c) $\frac{4}{9} \div \frac{2}{3} \times \frac{5}{8}$

2. Non-Permissible Values in a Division

In $\frac{A}{B} \div \frac{C}{D}$, the denominators B and D cannot be 0. The divisor $\frac{C}{D}$ cannot be 0 either, because division by zero is undefined; this happens when $C = 0$. After flipping, C is in a denominator, which is another way to see it.

NPVs of $\frac{A}{B} \div \frac{C}{D}$: every value that makes $B = 0$, $D = 0$ **or** $C = 0$ (the numerator of the divisor). Find them all from the original expression, before you flip.

Example 2

Simplify. State the non-permissible values.

(a) $\frac{12a^3}{5b^2} \div \frac{18a}{25b^5}$ (b) $\frac{-14x^2y}{9z^3} \div \frac{21xy^4}{6z}$

Example 3

State the non-permissible values. Do not simplify.

(a) $\frac{x+2}{x-5} \div \frac{x+7}{x}$ (b) $\frac{x^2-9}{x^2+4x} \div \frac{x^2-x-6}{2x-1}$

3. Dividing Rational Expressions with Polynomials

To divide rational expressions: (1) factor every numerator and denominator; (2) state the NPVs (both denominators and the divisor's numerator); (3) multiply by the reciprocal of the divisor; (4) divide out common factors and multiply. Leave the answer in factored form.

A polynomial divisor such as $x^2 + 3x - 10$ is $\frac{x^2+3x-10}{1}$, so its reciprocal is $\frac{1}{x^2+3x-10}$.

Example 4

Simplify. State the non-permissible values.

(a) $\dfrac{x+4}{x^2-2x} \div \dfrac{3x+12}{x^2-4}$

(b) $\dfrac{5y^2}{y^2-36} \div \dfrac{10y}{y+6}$

(c) $\dfrac{x^2-25}{4x} \div (x^2+3x-10)$

Example 5

Simplify. State the non-permissible values (restrictions).

(a) $\dfrac{x^2+x-12}{x^2-x-2} \div \dfrac{x^2+6x+8}{x^2-1}$

(b) $\dfrac{2a^2-a-6}{a^2-9} \div \dfrac{4a^2-9}{6-2a}$

(c) $\dfrac{x^2-y^2}{x^2+4xy+3y^2} \div \dfrac{2x^2-2xy}{x+3y}$

4. Mixed Multiplication and Division

Multiplication and division have equal priority, so work from **left to right**. The safe method: change each $\div$ into $\times$ by flipping *only the expression right after it*, then simplify the whole product at once. Note that $\dfrac{4}{9} \div \dfrac{2}{3} \times \dfrac{5}{8} = \dfrac{5}{12}$, but $\dfrac{4}{9} \div \left(\dfrac{2}{3} \times \dfrac{5}{8}\right) = \dfrac{16}{15}$: do not group the last two unless there are brackets. The NPVs come from every denominator and from the numerator of every divisor.

Example 6

Simplify. State the non-permissible values.

(a) $\dfrac{x^2-4x}{x+3} \div \dfrac{x^2-16}{x^2+3x} \cdot \dfrac{x+4}{2x}$

(b) $\dfrac{m^2+m-6}{m^2-4} \cdot \dfrac{m^2+5m+6}{m^2+6m+9} \div \dfrac{m+2}{3m}$

5. Complex Fractions and Problems

A **complex fraction** has a fraction in its numerator, its denominator or both. The main fraction bar means divide:

$$\frac{\frac{A}{B}}{\frac{C}{D}} = \frac{A}{B} \div \frac{C}{D}$$ NPVs: $B = 0$, $D = 0$ or $C = 0$.

(Complex fractions with sums or differences inside them need the adding and subtracting skills of Lessons 5 and 6.) In problems, division answers questions such as "width = area ÷ length" and "how many times larger".

Example 7

Simplify $\dfrac{\frac{x^2-9}{6x}}{\frac{x^2+x-12}{4x^2}}$. State the non-permissible values.

Example 8

A rectangle has area $\dfrac{x^2+5x}{x-2}$ square centimetres and length $\dfrac{x^2-25}{x^2-4}$ centimetres, where $x > 5$.

(a) Write and simplify an expression for its width.

(b) Determine the width when $x = 8$, to the nearest tenth.

Lesson 5: Adding and Subtracting with Monomial Denominators

Finding the lowest common denominator of numbers and monomials, writing equivalent fractions, and combining rational expressions into simplest form.

1. Review: Adding and Subtracting Fractions

Fractions can be added or subtracted only when their denominators are the same. The smallest denominator that works is the **lowest common denominator (LCD)**: the lowest common multiple of the denominators.

To add or subtract fractions: (1) find the LCD; (2) write each fraction as an equivalent fraction with the LCD as its denominator (multiply its numerator *and* denominator by the same number); (3) add or subtract the numerators and keep the denominator; (4) simplify.

A calculator's fraction template can check these answers, but it is the steps that carry over to rational expressions.

Example 1

Simplify.

(a) $\frac{5}{6}+\frac{3}{8}$ (b) $\frac{7}{10}-\frac{4}{15}$ (c) $\frac{2}{3}-\frac{5}{12}+\frac{3}{4}$ (d) $\frac{4}{9}-\frac{7}{6}$

2. Numerical Denominators

The same steps work when the numerators contain a variable. Multiply the *whole* numerator by the factor (use brackets), then expand and collect like terms. When subtracting, the minus sign applies to every term of the numerator that follows it.

$$\frac{a}{c}-\frac{b+d}{c}=\frac{a-(b+d)}{c}=\frac{a-b-d}{c}$$ Put each numerator in brackets before you subtract.

These denominators are numbers, never 0, so there are no non-permissible values.

Example 2

Simplify.

(a) $\frac{x+3}{4}+\frac{2x-1}{6}$ (b) $\frac{3a+2}{5}-\frac{a-4}{10}$ (c) $\frac{y-2}{3}-\frac{2y+5}{4}+\frac{y}{6}$

3. Expressions with the Same Denominator

When the denominators are already the same, add or subtract the numerators and keep the denominator. State the non-permissible values (NPVs) from the denominator first. Then simplify: factor the numerator and cancel common *factors*. The NPVs stay with the answer, even if the factor that caused them cancels.

Example 3

Simplify. State the non-permissible values.

(a) $\frac{7}{2x} + \frac{3}{2x}$ (b) $\frac{4a+1}{3a} - \frac{a-5}{3a}$ (c) $\frac{5m-2}{m+4} - \frac{2m-14}{m+4}$ (d) $\frac{x^2}{x-3} - \frac{9}{x-3}$

4. The LCD of Monomials

For monomial denominators, build the LCD one piece at a time.

LCD of monomials: the lowest common multiple of the coefficients, times each variable raised to the *highest* power that appears. e.g. $6x^2y$ and $4xy^3$: LCD $= 12x^2y^3$.

To rewrite an expression over the LCD, ask "what do I multiply my denominator by to get the LCD?" and multiply the numerator and the denominator by that monomial.

Example 4

Determine the LCD.

(a) $5x$ and $3x^2$ (b) $8a$ and $12b$ (c) $6m^2n$ and $9mn^3$ (d) $2x$, 4 and x^3

_____ _____ _____ _____

Fill in the missing numerator. (e) $\frac{5}{6x^2y} = \frac{\quad}{12x^2y^3}$ (f) $\frac{3}{4xy^3} = \frac{\quad}{12x^2y^3}$

5. Different Monomial Denominators

To add or subtract rational expressions: (1) state the NPVs (each variable in a monomial denominator $\neq 0$); (2) find the LCD; (3) rewrite each expression over the LCD; (4) combine the numerators (brackets when subtracting), expand and collect like terms; (5) simplify: factor the numerator if possible and cancel common factors.

Example 5

Simplify. State the non-permissible values.

(a) $\frac{2}{3x} + \frac{5}{6x}$ (b) $\frac{4}{a} - \frac{3}{a^2}$

(c) $\frac{5}{6m^2n} - \frac{7}{9mn^3}$ (d) $\frac{1}{x} + \frac{2}{3x^2} - \frac{1}{2x^3}$

Example 6

Simplify. State the non-permissible values.

(a) $\dfrac{x+2}{3x} + \dfrac{x-4}{2x}$

(b) $\dfrac{2y-1}{4y} - \dfrac{y-3}{6y^2}$

(c) $\dfrac{3a-b}{a^2b} - \dfrac{a-2b}{ab^2}$

(d) $\dfrac{x+6}{2x} - \dfrac{x+9}{3x}$

6. Whole-Number Terms, Common Errors and Problems

A term with no denominator has denominator 1: $3 = \dfrac{3}{1}$, so it is rewritten over the LCD like any other term. Three common errors: adding the denominators ($\dfrac{2}{x} + \dfrac{3}{y} \neq \dfrac{5}{x+y}$), forgetting the brackets when subtracting, and cancelling *terms* instead of factors. A quick test for an answer: substitute a number (not an NPV) into the question and into the answer. If the values differ, there is an error. (Equal values do not prove the answer is right, but they catch most mistakes.)

Example 7

(a) Simplify $3 - \dfrac{2}{x}$.

(b) Simplify $\dfrac{x}{4} - \dfrac{3}{x} + 1$.

(c) Kayla's work is shown. Find her error and correct it.

$$\frac{5}{2x} - \frac{x-3}{2x} = \frac{5-x-3}{2x} = \frac{2-x}{2x}$$

(d) Diego wrote $\dfrac{2}{x} + \dfrac{3}{y} = \dfrac{5}{x+y}$. Show that this is wrong by substituting $x = 1$ and $y = 1$, then simplify correctly.

Example 8

A hiker climbs a 9 km trail at an average speed of v km/h and comes down a longer 15 km trail at $2v$ km/h. (Time $= \dfrac{\text{distance}}{\text{speed}}$.)

(a) Write the total time as a single rational expression in simplest form.

(b) Determine the total time when $v = 3$ km/h.

(c) Write a simplified expression for how much longer the climb takes than the descent, and evaluate it for $v = 3$ km/h.

Lesson 6: Adding and Subtracting with Binomial and Trinomial Denominators

Factoring denominators to find the LCD, subtracting binomial numerators carefully, and simplifying the result fully.

1. Binomial Denominators with No Common Factor

The method is the same as for monomial denominators. When the denominators have no common factor, the LCD is their **product**. Leave the LCD in factored form: it shows the NPVs and makes the final check for common factors easy.

$\dfrac{a}{x+p} + \dfrac{b}{x+q} = \dfrac{a(x+q)+b(x+p)}{(x+p)(x+q)}$	Expand the numerator, collect like terms, then factor it to see whether anything cancels.

Example 1

Simplify. State the non-permissible values.

(a) $\dfrac{3}{x+2} + \dfrac{5}{x-4}$ (b) $\dfrac{4}{a-3} - \dfrac{1}{2a+1}$ (c) $\dfrac{2}{x} - \dfrac{3}{x+5}$

With binomial numerators, each product is a binomial times a binomial. **Expand the product being subtracted inside brackets first**, then remove the brackets, changing every sign.

Example 2

Simplify. State the non-permissible values.

(a) $\dfrac{x+1}{x-2} - \dfrac{x-3}{x+4}$ (b) $\dfrac{2y}{y+3} + \dfrac{y-1}{y-2}$ (c) $2 - \dfrac{x+1}{x+3}$

2. Factor First: Denominators with a Common Factor

If the denominators share a factor, multiplying them gives a common denominator that is too big, and the answer then needs extra simplifying. Factor every denominator first.

LCD of polynomials: factor each denominator fully. The LCD is the product of every different factor, each used the greatest number of times it appears in any one denominator. e.g. $x^2 - 4x = x(x-4)$ and x: LCD $= x(x-4)$.

Example 3

Simplify. State the non-permissible values.

(a) $\dfrac{5}{2x+6} - \dfrac{1}{x+3}$ (b) $\dfrac{3}{x^2-4x} + \dfrac{2}{x}$ (c) $\dfrac{4}{3a-6} - \dfrac{5}{4a-8}$

Always factor the numerator of the result. It may share a factor with the denominator, and the answer is not in simplest form until that factor is cancelled.

Example 4

Simplify fully. State the non-permissible values.

(a) $\dfrac{2}{x-1} - \dfrac{4}{x^2-1}$ (b) $\dfrac{x}{x-5} - \dfrac{25}{x^2-5x}$

3. Trinomial Denominators

Factor each trinomial (Unit 4), then build the LCD from the factors. Most trinomial questions share one factor, so the LCD has three factors, not four.

Example 5

Simplify fully. State the non-permissible values.

(a) $\dfrac{3}{x^2+5x+6}+\dfrac{2}{x^2-x-6}$

(b) $\dfrac{3}{x^2-2x-8}-\dfrac{1}{x^2-6x+8}$

(c) $\dfrac{2}{2x^2+5x-3}+\dfrac{1}{x^2+3x}$

4. Opposite Factors and Simplifying Each Term First

Factors such as $x-5$ and $5-x$ are **opposites**: $5-x=-(x-5)$. Rewrite the opposite factor and move its negative sign to the front of the fraction (this changes + to − or − to +).

$$\frac{b}{a-x}=\frac{b}{-(x-a)}=-\frac{b}{x-a}$$ e.g. $9-a^2=-(a^2-9)=-(a-3)(a+3)$

If one of the expressions can be simplified on its own, simplify it first, but take the NPVs from the *original* denominators.

Example 6

Simplify fully. State the non-permissible values.

(a) $\dfrac{6}{x-5}+\dfrac{2}{5-x}$

(b) $\dfrac{3x}{x^2-16}-\dfrac{2}{4-x}$

(c) $\dfrac{a+2}{a^2-a-6}+\dfrac{2}{9-a^2}$

5. Problems, Checking and an Extension

To check an answer, substitute a value that is not an NPV (such as $x = 1$) into the question and the answer, or enter the question as Y_1 and the answer as Y_2 on a graphing calculator and compare the TABLE values. In a problem, also state the values of the variable that make sense in the context.

Example 7

A boat travels 36 km up a river and 36 km back. Its speed in still water is 12 km/h and the current flows at c km/h, so the boat moves at $12 - c$ km/h upstream and $12 + c$ km/h downstream.

(a) Write the total travel time as a single rational expression in simplest form. What values of c make sense?

(b) Determine the total time when the current is 3 km/h, and check with the original expression.

(c) How much longer does the upstream trip take than the downstream trip? Write a simplified expression and evaluate it for $c = 3$.

Extension: complex fractions. A **complex fraction** has a fraction in its numerator or denominator. Simplify the top and the bottom into single fractions, then divide (multiply by the reciprocal, Lesson 4). The NPVs come from every denominator, and the whole divisor cannot be 0.

Example 8

Simplify $\dfrac{1 + \dfrac{1}{x}}{1 - \dfrac{1}{x^2}}$. State the non-permissible values.

Lesson 7: Solving Rational Equations

Stating non-permissible values, clearing the denominators with the LCD, and checking for roots that must be rejected.

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1. Rational Equations and Their Non-Permissible Values

A **rational equation** is an equation with a variable in a denominator, such as $\dfrac{3}{x} + 1 = \dfrac{5}{2x}$. Before solving, find the **non-permissible values** (NPVs): factor every denominator in the *original* equation and set each factor with a variable equal to 0. No NPV can be a solution, because it would make a denominator 0.

A value is a **solution** (root) if it is permissible and makes the left side (LS) equal to the right side (RS). To **verify** a solution, substitute it into the original equation and evaluate each side separately.

Example 1

State the non-permissible values.

(a) $\frac{7}{x} + 2 = \frac{3}{4x}$

(b) $\frac{x}{x+5} = \frac{2}{3x-1}$

(c) $\frac{1}{x^2-6x} = \frac{4}{x^2-36}$

(d) $\frac{3}{x^2+x-12} - \frac{1}{x} = 2$

Example 2

Verify whether the given value is a solution of the equation.

(a) $x = 3; \quad \frac{6}{x} + 1 = \frac{9}{x}$

(b) $x = 1; \quad \frac{x}{x+3} = \frac{4}{x}$

2. Solving by Multiplying by the LCD

Multiplying *every term on both sides* by the lowest common denominator (LCD) clears all the fractions and leaves a polynomial equation. Terms with no denominator (like the 1 in $\frac{3}{x} + 1$) must be multiplied too.

Solving a rational equation. (1) Factor each denominator and state the NPVs. (2) Multiply every term by the LCD and simplify. (3) Solve the resulting equation. (4) Reject any value that is an NPV. (5) Verify in the original equation.

Example 3

State the NPVs, then solve. Verify the solution to (a).

(a) $\frac{2}{x} + \frac{1}{3} = \frac{7}{3x}$

(b) $\frac{x+1}{2x} - \frac{3}{4} = \frac{1}{x}$

(c) $\frac{4}{3x} + \frac{x-2}{x} = \frac{5}{6}$

3. One Fraction on Each Side: Cross-Multiplying

An equation with exactly one fraction on each side is a **proportion**. Cross-multiplying is a shortcut for multiplying both sides by the LCD.

If $\frac{a}{b} = \frac{c}{d}$ $(b, d \neq 0)$, then $ad = bc$.

Put brackets around binomials when you cross-multiply. If x^2 terms appear on both sides, they may cancel and leave a linear equation.

Example 4

State the NPVs and solve.

(a) $\dfrac{4}{x-1} = \dfrac{3}{x+2}$ (b) $\dfrac{2x+1}{x-3} = \dfrac{4x-5}{2x+1}$ (c) $\dfrac{5}{3-y} = 2$

Example 5

A club has 7 more girls than boys. The ratio of boys to girls is $4:5$. Write and solve a rational equation to determine the number of boys and girls.

4. Binomial and Trinomial Denominators

Factor every denominator first. The LCD is the product of all the different factors, each used the greatest number of times it appears in any one denominator. Denominators that are opposites, like $x-4$ and $4-x$, are related by $4-x = -(x-4)$, so $\dfrac{3}{4-x} = -\dfrac{3}{x-4}$.

Example 6

State the NPVs, solve and verify.

(a) $\dfrac{3}{x+2}+\dfrac{1}{x-2}=\dfrac{8}{x^2-4}$

(b) $\dfrac{2}{x-1}-\dfrac{1}{x+3}=\dfrac{6}{x^2+2x-3}$

(c) $\dfrac{7}{x-4}+\dfrac{3}{4-x}=2$

(d) $\dfrac{3}{x}+\dfrac{x}{x+2}=1$

5. Rejected Values and Equations with No Solution

Multiplying by the LCD multiplies by an expression that equals 0 at each NPV, so the polynomial equation can have a root that the original equation does not. Such a root is **extraneous** and must be rejected. If every value you find is rejected, or the variable cancels and leaves a false statement (like $0 = 5$), the equation has **no solution**.

Example 7

Solve.

(a) $\dfrac{3x}{x-2}=1+\dfrac{6}{x-2}$

(b) $\dfrac{2}{x+3}+\dfrac{1}{x-3}=\dfrac{6}{x^2-9}$

6. Solving with a Graphing Calculator

The solutions of an equation are the x-coordinates of the points where the graphs of $y =$ LS and $y =$ RS intersect.

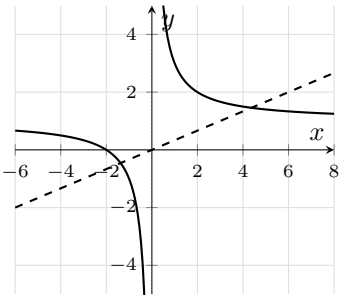
Intersect method: $Y_1 =$ left side, $Y_2 =$ right side; graph, then use *intersect* at each crossing.
Zero method: $Y_1 =$ left side − right side; graph, then use *zero* at each x-intercept.

Use brackets: type $\frac{5}{x+1}$ as `5/(X+1)`. Each NPV gives a vertical asymptote, where the graph breaks into separate pieces; the calculator may draw a near-vertical line there that is not part of the graph. Choose a window that shows every intersection.

Example 8

(a) Solve $\dfrac{5}{x+1} = \dfrac{2}{x-3}$ algebraically, then confirm with a graphing calculator.

(b) The graphs of $y = \dfrac{2}{x} + 1$ and $y = \dfrac{x}{3}$ are shown. Use a graphing calculator to solve $\dfrac{2}{x} + 1 = \dfrac{x}{3}$, to the nearest hundredth. State a suitable window.



solid: $y = \frac{2}{x} + 1$ dashed: $y = \frac{x}{3}$

Lesson 8: Rational Equations That Lead to Quadratics

Clearing denominators to get a quadratic equation, solving it by factoring or the quadratic formula, and rejecting non-permissible roots.

1. When the LCD Gives a Quadratic Equation

Often, after multiplying by the LCD, x^2 terms do not cancel and the result is a **quadratic equation**. Write it in standard form $ax^2 + bx + c = 0$, then solve it by factoring (or by the quadratic formula). A quadratic gives up to two roots, and *each one* must be compared with the NPVs and checked.

(1) Factor the denominators; state the NPVs. (2) Multiply every term by the LCD. (3) Write the quadratic in standard form. (4) Solve by factoring or the quadratic formula. (5) Reject any root that is an NPV. (6) Verify the remaining roots.

Example 1

State the NPVs, solve and verify.

(a) $x + \dfrac{6}{x} = 5$ (b) $\dfrac{x}{x-2} = \dfrac{8}{x}$ (c) $\dfrac{3}{x} + \dfrac{x}{x+2} = 2$

2. Rejecting a Non-Permissible Root

When one root of the quadratic is an NPV, it is **extraneous**: it solves the polynomial equation but makes a denominator of the original equation 0. Reject it and keep the other root. If both roots are rejected, the equation has **no solution**.

Example 2

Consider the equation $\dfrac{x^2-2x-15}{x-5}=2x$.

(a) State the NPV and solve by cross-multiplying.

(b) Explain why $x=5$ does not satisfy the original equation, even though it solves the quadratic.

Example 3

State the NPVs and solve.

(a) $\dfrac{x}{x-3}-\dfrac{2}{x+1}=\dfrac{8}{x^2-2x-3}$

(b) $\dfrac{x+1}{x-4}=\dfrac{5}{4-x}+x$

Example 4

Show that $\dfrac{x}{x-3}-\dfrac{3}{x}=\dfrac{9}{x^2-3x}$ has no solution.

3. Using the Quadratic Formula

If the quadratic does not factor, use the quadratic formula. Give exact roots in simplest radical form unless the question asks for a rounded answer. A graphing calculator (graph both sides and use intersect) confirms the decimal values.

For $ax^2+bx+c=0$: $\quad x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$. $\quad$ If $b^2-4ac<0$, there are no real roots, so the rational equation has no real solution.

Example 5

Solve. Give exact answers, then round to the nearest hundredth.

(a) $\frac{2}{x} + 1 = \frac{x}{3}$ (solved graphically in Lesson 7)

(b) $\frac{1}{x-1} + \frac{1}{x+1} = 1$

Example 6

Solve.

(a) $\frac{x+2}{x-1} = \frac{2x+1}{x+1}$

(b) $\frac{x+4}{2x-1} = \frac{x+1}{x-1}$

4. Rational Equations in Formulas

Many science formulas contain reciprocals. When two resistors R_1 and R_2 (in ohms, Ω) are connected in parallel, the combined resistance R satisfies

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

Substituting the given information often gives a rational equation that leads to a quadratic. Reject a root that is not reasonable (a resistance is positive).

Example 7

Two resistors are connected in parallel. One has a resistance 5 Ω greater than the other, and their combined resistance is 6 Ω. Determine the resistance of each.

Lesson 9: Problems Involving Rational Equations (work, motion, numbers)

Modelling number, distance–speed–time and work-rate situations with a rational equation, solving it and rejecting roots that do not fit the situation.

1. A Plan, and Number Problems

Many problems lead to an equation with the variable in a denominator: a reciprocal, a time $\dfrac{\text{distance}}{\text{speed}}$, or a rate of work $\dfrac{1}{\text{time}}$.

Solving a problem with a rational equation. (1) Let a variable stand for one unknown and write the other unknowns in terms of it (a table helps). (2) Write the equation from the sentence that links the quantities. (3) State the non-permissible values, multiply every term by the LCD and solve (factor, or use the quadratic formula and round as asked). (4) **Reject inadmissible roots**: NPVs, negative speeds, times or lengths, and non-integers when the answer must be an integer. (5) Check in the words of the problem and answer the question with units.

The **reciprocal** of a number n ($n \neq 0$) is $\dfrac{1}{n}$. Consecutive integers are $n, n+1, \ldots$; consecutive odd (or even) integers are $n, n+2, \ldots$.

Example 1

(a) The sum of a number and twice its reciprocal is $\dfrac{11}{3}$. Determine the number.

(b) The sum of the reciprocals of two consecutive odd integers is $\dfrac{12}{35}$. Determine the integers.

(c) The denominator of a fraction is 5 more than its numerator. When 3 is added to the numerator and to the denominator, the new fraction is equal to $\dfrac{3}{4}$. Determine the original fraction.

2. Distance, Speed and Time

$d = st \qquad t = \dfrac{d}{s} \qquad s = \dfrac{d}{t}$ (distance d, average speed s, time t). Make a table with a row for each trip and the columns Distance, Speed, Time $= \dfrac{\text{Distance}}{\text{Speed}}$. Then use the time sentence: **same time** $\Rightarrow$ the times are equal; k **hours longer** $\Rightarrow$ longer time $-$ shorter time $= k$; **total time** $\Rightarrow$ add the times. Change minutes to hours: 40 min $= \frac{40}{60} = \frac{2}{3}$ h.

Example 2

A cyclist rides 54 km in the same time that a runner covers 24 km. The cyclist's average speed is 15 km/h greater than the runner's. Determine each average speed.

	Distance (km)	Speed (km/h)	Time (h)
Cyclist			
Runner			

Example 3

A courier drives 160 km to a town. On the way back she drives 20 km/h faster and saves 40 minutes. Determine her speed each way and the time for the return trip.

	Distance (km)	Speed (km/h)	Time (h)
Going			
Returning			

Example 4

In a duathlon, Mara cycles 15 km and then runs 6 km. Her running speed is 12 km/h less than her cycling speed, and the race takes her 1.5 h. Determine her cycling and running speeds.

3. Current and Wind

A current or a wind adds to the speed in one direction and subtracts from it in the other.

Boat with still-water speed s, current c: **downstream** speed $s + c$, **upstream** speed $s - c$ $(s > c)$.
Plane with airspeed s, wind w: with a **tailwind** $s + w$, into a **headwind** $s - w$.

Example 5

A tour boat travels 24 km upstream and then 24 km back downstream in a total of 5 h. The current flows at 2 km/h. Determine the speed of the boat in still water and the time for each leg.

	Distance (km)	Speed (km/h)	Time (h)
Upstream			
Downstream			

4. Work Rate

If a job takes 6 h alone, the worker's **rate** is $\frac{1}{6}$ of the job per hour, and in t hours the worker does $\frac{t}{6}$ of the job. The parts done by everyone add to 1 (one whole job).

Times alone a and b, working together for t hours: $\frac{t}{a}+\frac{t}{b}=1$ (or $\frac{1}{a}+\frac{1}{b}=\frac{1}{t}$).

A drain or anything that undoes the work subtracts: $\frac{t}{a}-\frac{t}{b}=1$. If people work for different times, each part is $\frac{\text{time that person works}}{\text{time that person needs alone}}$.

Example 6

(a) Ari can paint a fence in 6 h and Bea can paint it in 4 h. How long will it take them working together?

(b) A large pump empties a flooded basement in 3 h. With a small pump also working, the job takes 2 h. How long would the small pump take alone?

(c) An inlet pipe fills a pool in 8 h and the drain empties it in 12 h. The drain is left open by mistake while the pool fills. How long does it take to fill?

(d) Dev can stain a deck in 10 h and Eli in 15 h. Dev works alone for 5 h, then Eli joins him. How long do they work together to finish?

Example 7

(a) Working alone, Jordan takes 5 h longer than Kai to clear the snow from an outdoor rink. Together they take 6 h. How long does each take alone?

(b) A new printer finishes a job 2 h faster than an old one. Together they finish it in 3 h. To the nearest tenth of an hour, how long does each take alone?

5. Other Rational Models

A ratio or percent that changes ($\frac{\text{part}}{\text{whole}}$ with the same amount added to both), or a given formula with reciprocals, also leads to a rational equation.

Example 8

(a) A basketball player has made 14 of her 25 free throws this season. How many free throws in a row must she make to raise her success rate to 75%?

(b) For a lens with focal length f, an object at distance u forms an image at distance v, where $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$. A lens has $f = 4$ cm and the image is 6 cm farther from the lens than the object. Determine u and v.

Lesson 10: Unit 6 Review

Non-permissible values, simplifying, the four operations, rational equations and their problems on one page, mixed examples that combine them, and a review assignment written like the unit test.

1. Unit Summary

Skill	What to do	Example
Non-permissible values (NPVs)	Factor each denominator; the values that make it 0 are NPVs. State them from the *original* expression.	$\dfrac{x+1}{x^2-3x-10}$: $(x-5)(x+2)$, $x \neq 5, -2$
Simplifying	Factor, state NPVs, divide out common **factors** (never terms). $\dfrac{a-b}{b-a} = -1$.	$\dfrac{x^2-2x-8}{x^2-16} = \dfrac{(x-4)(x+2)}{(x-4)(x+4)} = \dfrac{x+2}{x+4}$, $x \neq \pm 4$
Multiplying	Factor everything, state NPVs, divide out factors common to any numerator and any denominator, multiply.	$\dfrac{x^2-4}{3x} \cdot \dfrac{6x^2}{x+2} = 2x(x-2)$, $x \neq 0, -2$
Dividing	Multiply by the reciprocal of the divisor. NPVs: all original denominators **and** the divisor's numerator.	$\dfrac{x}{x-1} \div \dfrac{x+3}{x-1} = \dfrac{x}{x+3}$, $x \neq 1, -3$
Adding and subtracting	Factor denominators, find the LCD, rewrite each fraction over the LCD, combine numerators (brackets when subtracting), simplify.	$\dfrac{3}{x} - \dfrac{2}{x+1} = \dfrac{3(x+1)-2x}{x(x+1)} = \dfrac{x+3}{x(x+1)}$, $x \neq 0, -1$
Solving equations	State NPVs, multiply every term by the LCD, solve (linear or quadratic), reject any root that is an NPV (**extraneous**), verify.	$\dfrac{x}{x-2} = \dfrac{2}{x-2} + 3$, $x \neq 2$: $x = 2 + 3(x-2)$ gives $x = 2$, an NPV: **no solution**
Problems	Table; $t = \dfrac{d}{s}$; downstream $s+c$, upstream $s-c$; work: $\dfrac{t}{a} + \dfrac{t}{b} = 1$. Reject inadmissible roots.	Same time: $\dfrac{d_1}{s_1} = \dfrac{d_2}{s_2}$; k h longer: $t_{\text{long}} - t_{\text{short}} = k$

Simplest form checklist. (1) Every numerator and denominator factored fully (GCF first, then trinomial or difference of squares). (2) No factor common to the numerator and the denominator (look for opposites like $3-x$ and $x-3$). (3) Like denominators combined into one fraction. (4) NPVs stated from the original expression, including any divisor's numerator. The simplified expression is **equivalent** to the original for every permissible value.

Rational equations. An equation is solved when the variable is out of the denominators: multiply *every* term (including terms with no denominator) by the LCD. A root that is a non-permissible value would make a denominator 0, so it is **extraneous**; if every root is extraneous, there is **no solution**.

Common errors. $\dfrac{x+6}{x+3} \neq 2$ (the x's are terms, not factors). NPVs taken from the simplified answer miss the cancelled factors. $\dfrac{a}{b} - \dfrac{c+d}{b} = \dfrac{a-c-d}{b}$, not $\dfrac{a-c+d}{b}$. Forgetting to multiply a constant term by the LCD.

2. Expressions

Example 1

State the non-permissible values, then simplify.

(a) $\dfrac{5x}{2x^2+7x-4}$ (NPVs only)

(b) $\dfrac{3x^2-12x}{x^2-x-12}$

(c) $\dfrac{10-5x}{x^2-4}$

(d) $\dfrac{x^2-9y^2}{x^2+xy-6y^2}$ (NPVs in terms of y)

Example 2

Simplify and state the non-permissible values.

(a) $\dfrac{x^2-25}{4x+8}\cdot\dfrac{x^2+2x}{x^2-3x-10}$

(b) $\dfrac{2x^2+5x-3}{x^2-9}\div\dfrac{4x^2-1}{x^2-3x}$

Example 3

Simplify and state the non-permissible values.

(a) $\dfrac{5}{6x^2}-\dfrac{3}{4x}+\dfrac{1}{3}$

(b) $\dfrac{2x}{x^2-4}-\dfrac{1}{x-2}$

Example 4

(a) Simplify $\left(\frac{1}{x}+\frac{1}{2}\right) \div \frac{x+2}{x^2}$ and state the non-permissible values.

(b) A rectangle has area $\frac{3x^2+10x-8}{x^2-4}$ and length $\frac{3x-2}{x-2}$. Write its width in simplest form, then find the length and width when $x = 6$.

3. Equations and Problems

Example 5

State the non-permissible values, solve and verify.

(a) $\frac{3}{x+2} = \frac{5}{2x-1}$

(b) $\frac{x}{x-3} - \frac{18}{x^2-9} = \frac{4}{x+3}$

Example 6

(a) Show that $\frac{2x}{x-4} = 3 + \frac{8}{x-4}$ has no solution.

(b) Explain why a root of the equation you get after multiplying by the LCD may not be a root of the original equation.

Example 7

A train travels 540 km. A bus that is 30 km/h slower takes 1.5 h longer to cover the same distance. Determine both speeds and both times.

Example 8

Working alone, one snowplow takes 4 h longer than another to clear a parking lot. Together they clear it in 4 h 48 min. How long does each take alone?